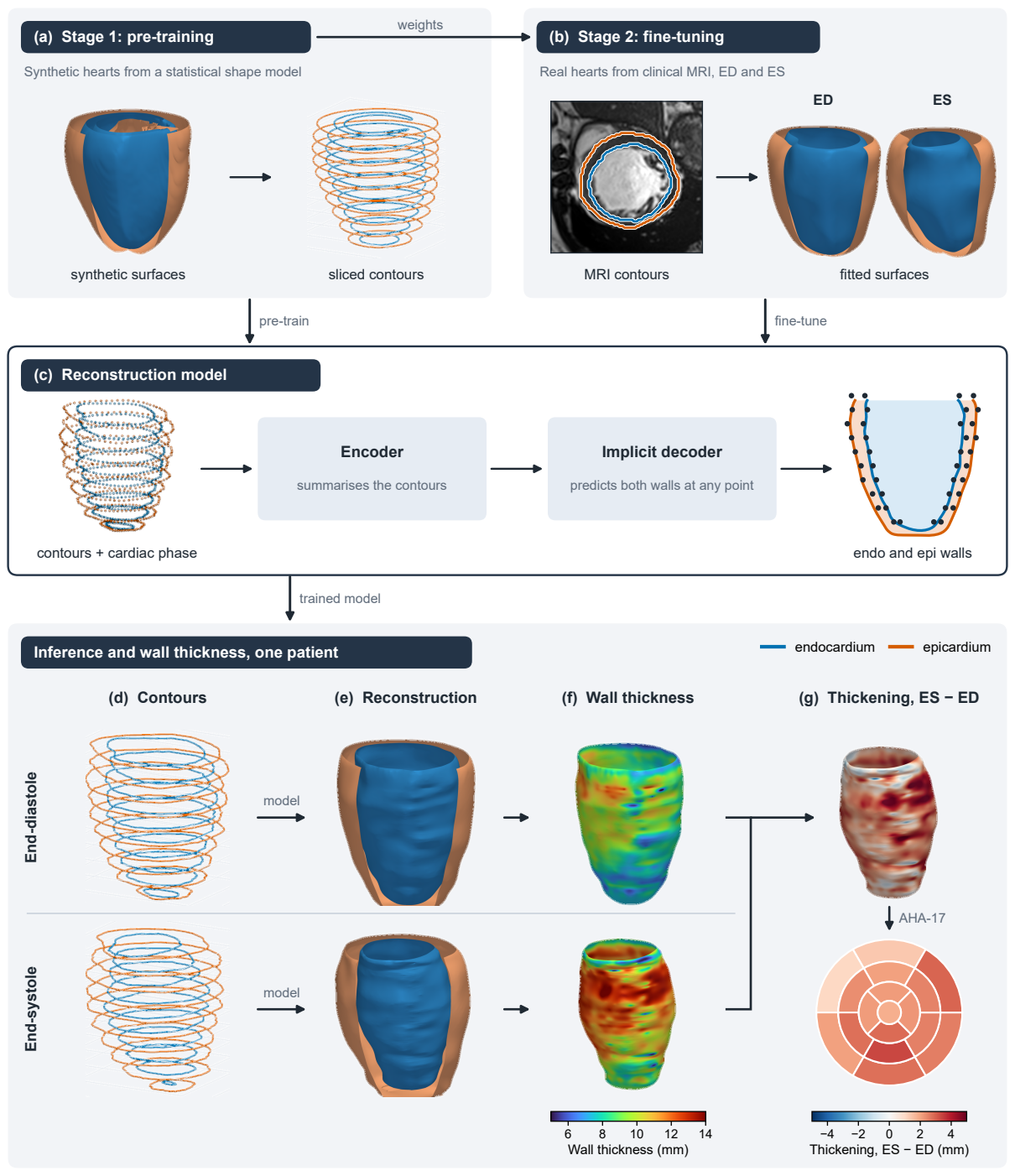

(a) Stage 1: pre-training
weights
(b) Stage 2: fine-tuning
Synthetic hearts from a statistical shape model
Real hearts from clinical MRI, ED and ES
ED
ES
synthetic surfaces
sliced contours
MRI contours
fitted surfaces
pre-train
fine-tune
(c) Reconstruction model
Encoder
summarises the contours
Implicit decoder
predicts both walls at any point
contours + cardiac phase
endo and epi walls
trained model
Inference and wall thickness, one patient
endocardium
epicardium
(d) Contours
(e) Reconstruction
(f) Wall thickness
(g) Thickening, ES − ED
End-diastole
End-systole
model
model
AHA-17
6 8 10 12 14
Wall thickness (mm)
−4 −2 0 2 4
Thickening, ES − ED (mm)